Sound Speed and Probe Misalignment Estimation for Diverging-Wave Imaging of Pipes

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***Abstract*—Ultrasound imaging is widely used in nondestructive testing (NDT) of various industrial structures, e.g., inspecting liquid-filled pipelines to assess their structural integrity. In this context, our work pertains to receive beamforming of diverging-wave imaging data acquired by an immersed transducer array probe positioned inside a pipe. We present a simple and effective method for estimating the speed-of-sound values and probe positioning errors, which is needed for calculating signal travel-time delays with higher accuracy, leading to better-quality multimode images produced by a delay-and-sum beamformer. Our imaging results demonstrate the importance of sound speed and probe misalignment estimation based on experimental data acquired by a 256-element probe.**

***Index Terms*—ultrasonic NDT, ultrafast imaging, speed-of-sound estimation, probe positioning errors.**

I. Introduction

High-frame-rate ultrasound imaging techniques have shown to be highly successful in both biomedical and nondestructive testing (NDT) fields [1]–[3]. The application context of this work is high-speed in-line inspection of liquid-filled large-diameter pipes using diverging waves (DW), which allows for fast data acquisition [4]. This setup involves a special tool with mounted transducer arrays moving along the pipe's interior. At every step, these arrays emit several DW pulses at various steering angles and record backscattered signals, used to reconstruct multimode images of a tested pipe section. To improve the accuracy of image reconstruction (e.g., using delay-and-sum beamforming [5]), we want to determine the actual speed-of-sound (SoS) values and probe positioning errors, as opposed to relying on the ideal-case nominal assumptions.

Fig. 1 illustrates the data acquisition geometry. The center of the transducer array, point O, serves as the origin of our (z, x) coordinate system. This probe is immersed in liquid inside a pipe whose frontwall (inner) and backwall (outer) radii are denoted by R_{fw} and R_{bw}, respectively. Ideal probe positioning would be centered at point O′, vertically aligned with the pipe center C. The difference between O′ and O (the actual probe center) is quantified by the axial and lateral offsets z_{err} and x_{err} that characterize probe misalignment.

Our objective is to find z_{err} and x_{err}, along with the SoS in liquid, denoted by c_l, as well as the longitudinal and transverse SoS in steel, denoted by $c_{s\mathrm{L}}$ and $c_{s\mathrm{T}}$. To accomplish this, we utilize direct-mode limited-range trial images reconstructed from three acquired channel data frames after DW emissions at zero, positive, and negative steering angles (e.g., $0°$, $20°$, $-20°$). Once the actual SoS values and the probe positioning errors are found, final reconstruction of complete multimode images can proceed using more accurate propagation delays that depend on c_l, $c_{s\mathrm{L}}$, $c_{s\mathrm{T}}$, z_{err}, and x_{err}.

Speed-of-sound estimation has been extensively investigated for aberration correction in biomedical imaging (e.g., see [6]). Also, various methods for determining the shape of inspected component surfaces have been reported in the NDT literature (e.g., see [7]). We pursue similar tasks and provide simplified image-based solutions specifically for the problem at hand.

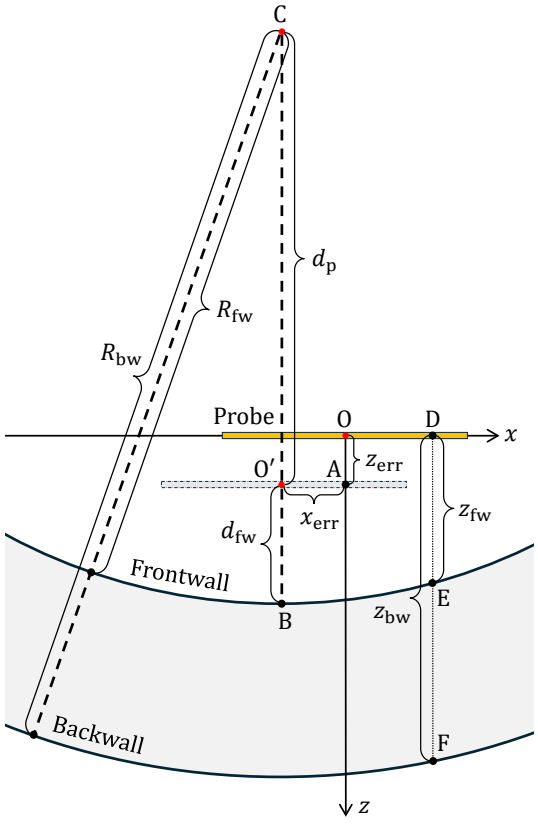


Fig. 1. Basic data acquisition geometry (not to scale).

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II. Proposed Method

Let $\mathcal{A}_0$, $\mathcal{A}_+$, and $\mathcal{A}_-$ represent channel data frames acquired by the probe after DW emissions at zero, positive, and negative steering angles, respectively. Given these inputs, our method performs the following steps:

1) Find c_l (SoS in liquid) using trial images of the frontwall section reconstructed from $\mathcal{A}_0$, $\mathcal{A}_+$, and $\mathcal{A}_-$.
2) Compute z_{err} and x_{err} (probe misalignment offsets) based on the frontwall images reconstructed using estimated c_l.
3) Determine $c_{s\text{L}}$ and $c_{s\text{T}}$ (longitudinal and transverse SoS in steel) using trial images of the backwall section reconstructed from $\mathcal{A}_0$, $\mathcal{A}_+$, and $\mathcal{A}_-$, which utilizes estimated c_l, z_{err}, and x_{err}.

The details of each step are provided below. For reconstructed images mentioned in this section, $z^{\min}$ and $z^{\max}$ refer to their vertical (depth) limits, while $x^{\min}$ and $x^{\max}$ refer to the horizontal (lateral) limits. The pixel size is specified by Δz and Δx. In all cases, we set $\Delta z = \Delta x$ and $[x^{\min}, x^{\max}] = [-L/2, L/2]$, where L is the lateral width of a transducer array.

A. Speed-of-Sound Estimation for Liquid Layer

Our goal is to obtain an estimate of c_l within some predetermined range $[c_l^{\min}, c_l^{\max}]$. We consider several trial SoS values $c_l^* = c_l^{\min}, c_l^{\min} + \Delta c_l, c_l^{\min} + 2\Delta c_l, \ldots$ up to $c_l^{\max}$, using some step size Δc_l. For each c_l^*, we apply IQ delay-and-sum (DAS) beamforming to $\mathcal{A}_0$, $\mathcal{A}_+$, and $\mathcal{A}_-$, which yields reconstructed images denoted by $\mathcal{I}_0^*$, $\mathcal{I}_+^*$, and $\mathcal{I}_-^*$. The vertical span of these images is limited to

$$z^{\min} = \sqrt{R_{\text{fw}}^2 - (L/2)^2} - d_{\text{p}} - d_{\text{extra}}, \quad z^{\max} = d_{\text{fw}} + d_{\text{extra}}. \tag{1}$$

where d_{p} and d_{fw} are the lengths of line segments $\mathtt{O'B}$ and $\mathtt{O'C}$ shown in Fig. 1. When $d_{\text{extra}} = 0$, the corresponding $z^{\max}$ and $z^{\min}$ are the z-coordinates of the respective farthest and nearest frontwall points at $x = 0$ and $x = \pm L/2$, assuming the ideal probe positioning centered at $\mathtt{O'}$. We set $d_{\text{extra}} = d_{\text{fw}}/4$ to capture additional pixels surrounding the frontwall boundary and to accommodate for potential probe misalignment.

After DAS beamforming with a given c_l^* value, the resulting trial images $\mathcal{I}_0^*$, $\mathcal{I}_+^*$, and $\mathcal{I}_-^*$ are converted to their vectorized forms $\mathbf{v}_0^*$, $\mathbf{v}_+^*$, and $\mathbf{v}_-^*$. Then, we compute the average *cosine similarity* (CS) between $\mathbf{v}_0^*$ and $\mathbf{v}_+^*$ and between $\mathbf{v}_0^*$ and $\mathbf{v}_-^*$:

$$\text{CS}(c_l^*) = \frac{1}{2}\left(\frac{\mathbf{v}_0^* \cdot \mathbf{v}_+^*}{\|\mathbf{v}_0^*\| \cdot \|\mathbf{v}_+^*\|} + \frac{\mathbf{v}_0^* \cdot \mathbf{v}_-^*}{\|\mathbf{v}_0^*\| \cdot \|\mathbf{v}_-^*\|}\right). \tag{2}$$

Given $\text{CS}(c_l^*)$ over all c_l^* of interest, we select the SoS value that corresponds to the cosine similarity maximum [8]:

$$c_l = \arg\max_{c_l^*} \{\text{CS}(c_l^*) \mid c_l^{\min} \le c_l^* \le c_l^{\max}\}. \tag{3}$$

B. Probe Misalignment Estimation

Once c_l has been estimated, the next step is to find z_{err} and x_{err}. They are needed to determine the frontwall boundary at which the propagating wavefront refracts between the liquid and steel layers. The coordinates (z_{fw}, x) of this boundary are related as follows:

$$z_{\text{fw}}(x, x_{\text{err}}, z_{\text{err}}) = \sqrt{R_{\text{fw}}^2 - (x + x_{\text{err}})^2} - d_{\text{p}} + z_{\text{err}}. \tag{4}$$

The above equation serves as a regression model (parameterized by z_{err} and x_{err}) that we fit to our data points $z_{\text{fw}}^*(x)$ over all x-coordinates ranging from $x^{\min}$ to $x^{\max}$.

To obtain z_{fw}^* for model fitting, we first reconstruct images $\mathcal{I}_0$, $\mathcal{I}_+$, and $\mathcal{I}_-$ using estimated c_l from the previous step. Their vertical span is limited to $[z^{\min}, z^{\max}]$ given by (1), which contains the frontwall section. Next, we coherently compound these images and compute the resulting 2D envelope. Then, for each x, we find the z-coordinate of the maximum-intensity pixel (attributed to the frontwall reflection), which gives sought data point $z_{\text{fw}}^*(x)$.

Finally, we fit extracted $z_{\text{fw}}^*(x)$ to the model equation given by (4). The output is z_{err} and x_{err} estimates that minimize the weighted least squares (LS) loss:

$$\text{LS}(x_{\text{err}}, z_{\text{err}}) = \sum_x w_x \left(z_{\text{fw}}^*(x) - z_{\text{fw}}(x, x_{\text{err}}, z_{\text{err}})\right)^2, \tag{5}$$

where w_x refers to the residual-dependent Huber weights [9].

C. Speed-of-Sound Estimation for Steel Layer

We want to estimate $c_{s\text{L}}$ and $c_{s\text{T}}$ restricted to their respective numerical ranges $[c_{s\text{L}}^{\min}, c_{s\text{L}}^{\max}]$ and $[c_{s\text{T}}^{\min}, c_{s\text{T}}^{\max}]$. Following a similar approach outlined in Section II-A, we examine several trial SoS values $c_{s\text{L}}^* = c_{s\text{L}}^{\min}, c_{s\text{L}}^{\min} + \Delta c_{s\text{L}}, c_{s\text{L}}^{\min} + 2\Delta c_{s\text{L}}, \ldots$ up to $c_{s\text{L}}^{\max}$ and $c_{s\text{T}}^* = c_{s\text{T}}^{\min}, c_{s\text{T}}^{\min} + \Delta c_{s\text{T}}, c_{s\text{T}}^{\min} + 2\Delta c_{s\text{T}}, \ldots$ up to $c_{s\text{T}}^{\max}$.

For each $c_{s\text{L}}^*$, we reconstruct an L-L mode[1] image $\mathcal{I}_0^*$ from zero-angle channel data $\mathcal{A}_0$ using an IQ DAS beamformer that takes into account estimated c_l, z_{err}, and x_{err} when calculating propagation delays inside and outside the steel layer. The depth range of $\mathcal{I}_0^*$ is restricted to

$$z^{\min} = \min_x\{z_{\text{bw}}(x)\} - d_{\text{extra}}, \quad z^{\max} = \max_x\{z_{\text{bw}}(x)\} + d_{\text{extra}},$$
$$z_{\text{bw}}(x) = \sqrt{R_{\text{bw}}^2 - (x + x_{\text{err}})^2} - d_{\text{p}} + z_{\text{err}}, \tag{6}$$

where $z_{\text{bw}}(x)$ gives the z-coordinates of the backwall boundary. We set $d_{\text{extra}} = 2c_{s\text{L}}^{\max}/f_c$, where f_c is the central frequency of the emitted DW pulse.

Next, we obtain the corresponding 2D envelope $\mathcal{E}^*(z, x) = |\mathcal{I}_0^*|$ and normalize it to the $[0, 1]$ range as follows:

$$\mathcal{E}_{\text{norm}}^*(z, x) = \frac{\mathcal{E}^*(z, x) - \min_{z,x}\{\mathcal{E}^*(z, x)\}}{\max_{z,x}\{\mathcal{E}^*(z, x)\} - \min_{z,x}\{\mathcal{E}^*(z, x)\}}. \tag{7}$$

This normalized envelope is then compared with a 2D binary template given by

$$\mathcal{B}(z, x) = \begin{cases} 1, & \text{if } |z - z_{\text{bw}}(x)| \le d_{\text{extra}}/4 \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

[1]The hyphen indicates an interaction of a propagating wave with a target, which essentially separates the transmit and receive signal paths. Symbol L refers to the longitudinal wave propagation. In the sequel, we also mention T-T and TT-T modes, where symbol T refers to the transverse (shear) wave propagation. The TT abbreviation means that a transverse wave reflects from the backwall (without mode conversion) before reaching the target.

In other words, $\mathcal{B}(z,x)$ equals 1 in the vicinity of the backwall boundary (within $\pm d_{\text{extra}}/4$ along the z-axis) and 0 otherwise.

The similarity between $\mathcal{E}^*_{\text{norm}}$ and $\mathcal{B}$ is quantified using the absolute error (AE) computed over all pixels:

$$\text{AE}(c^*_{s\text{L}}) = \sum_{z,x} |\mathcal{E}^*_{\text{norm}}(z,x) - \mathcal{B}(z,x)|. \tag{9}$$

Given $\text{AE}(c^*_{s\text{L}})$ over all $c^*_{s\text{L}}$ of interest, we select the SoS value that corresponds to the minimum absolute error:

$$c_{s\text{L}} = \arg\min_{c^*_{s\text{L}}} \{\text{AE}(c^*_{s\text{L}}) \mid c^{\text{min}}_{s\text{L}} \le c^*_{s\text{L}} \le c^{\text{max}}_{s\text{L}}\}. \tag{10}$$

We use the same approach to estimate $c_{s\text{T}}$, except that we now rely on T-T mode images $\mathcal{I}^*_+$ and $\mathcal{I}^*_-$ reconstructed from $\mathcal{A}_+$ and $\mathcal{A}_-$ for each trial $c^*_{s\text{T}}$ under consideration. Here, we let $d_{\text{extra}} = 2c^{\text{max}}_{s\text{T}}/f_c$ and $\mathcal{E}^*(z,x) = |\mathcal{I}^*_+ + \mathcal{I}^*_-|$. The latter is then normalized according to (7) and compared with $\mathcal{B}(z,x)$ in terms of the overall AE given by the right-hand side of (9). In the end, we choose

$$c_{s\text{T}} = \arg\min_{c^*_{s\text{T}}} \{\text{AE}(c^*_{s\text{T}}) \mid c^{\text{min}}_{s\text{T}} \le c^*_{s\text{T}} \le c^{\text{max}}_{s\text{T}}\}. \tag{11}$$

III. Evaluation Results

Our evaluation results are based on three pipe samples with manufactured defects mimicking surface-breaking backwall cracks (see Fig. 2). We refer to these samples as P_1, P_2, and P_3. They were immersed in water and insonified with a 256-element linear probe (0.3-mm pitch) utilizing three 4.0-MHz DW emissions angled at $\theta = 0°, 20°, -20°$. Table I provides additional experimental parameters relevant to this work.

TABLE I
Experimental Parameters

Parameter	Value
256-element probe width L	76.5 mm
DW emission angles θ	$0°, 20°, -20°$
DW central frequency f_c	4.0 MHz
Nominal standoff d_{fw}	20.0 mm
Pipe inner radius R_{fw}	193.7 mm
Pipe outer radius R_{bw}	203.2 mm
Nominal SoS in liquid c^{nom}_l	1480 m/s
Nominal transverse SoS in steel $c^{\text{nom}}_{s\text{T}}$	3120 m/s
Nominal longitudinal SoS in steel $c^{\text{nom}}_{s\text{L}}$	5640 m/s

During image reconstructions with an IQ DAS beamformer, we set $\Delta z = \Delta x = 0.1$ mm and $[x^{\text{min}}, x^{\text{max}}] = [-L/2, L/2]$, where $L = 76.5$ mm (probe width). During SoS estimation, the initial step sizes were the same in all cases: $\Delta c_l = \Delta c_{s\text{L}} = \Delta c_{s\text{T}} = 10$ m/s. However, before selecting our c_l, $c_{s\text{L}}$, and $c_{s\text{T}}$, the respective decision curves $\text{CS}(c^*_l)$, $\text{AE}(c^*_{s\text{L}})$, and $\text{AE}(c^*_{s\text{T}})$ were interpolated to the SoS step size of 1 m/s. The SoS search range covered about $\pm 5\%$ deviations from the nominal values (see Table I). For example, given $c^{\text{nom}}_l = 1480$ m/s, we used $[c^{\text{min}}_l, c^{\text{max}}_l] = [1400, 1560]$ m/s.

Table II summarizes our SoS estimation results along with fitted z_{err} and x_{err} for all three samples. The reported probe positioning errors arise from the frontwall model fit illustrated

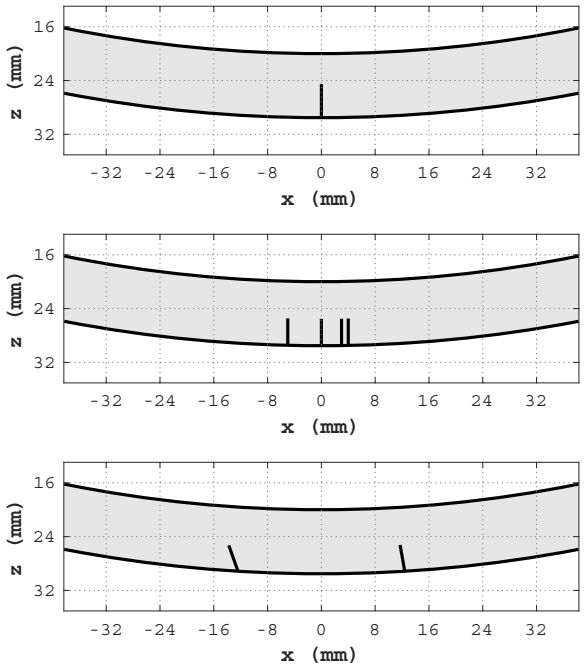


Fig. 2. Visualization of pipe samples P_1 (top), P_2 (middle), and P_3 (bottom).

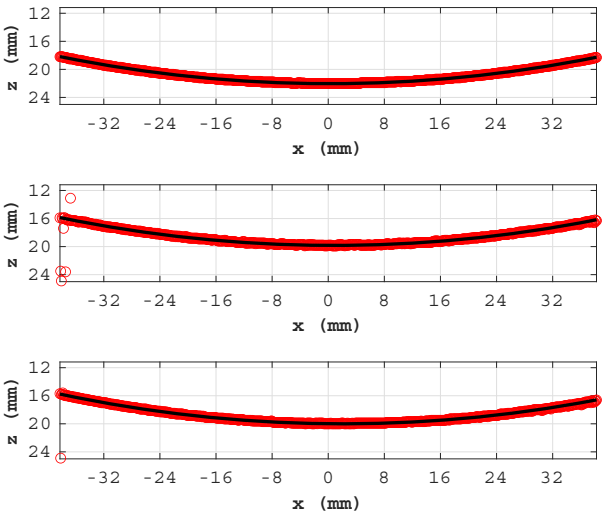


Fig. 3. Frontwall model fit for P_1 (top), P_2 (middle), and P_3 (bottom).

in Fig. 3, where circles represent input data points $z^*_{\text{fw}}(x)$, and solid curve is the model output $z_{\text{fw}}(x, x_{\text{err}}, z_{\text{err}})$. For sample P_1, Fig. 4 shows the plots of $\text{CS}(c^*_l)$, $\text{AE}(c^*_{s\text{L}})$, and $\text{AE}(c^*_{s\text{T}})$ over their respective SoS search ranges. Recall that our chosen c_l corresponds to the CS maximum, while $c_{s\text{L}}$ and $c_{s\text{T}}$ correspond to the AE minimum. Similar plots for samples P_2 and P_3 are shown in Fig. 5 and 6.

To illustrate the importance of speed-of-sound and probe misalignment estimation, Fig. 7–9 display illustrative DAS-beamformed images for two scenarios: 1) assuming the nominal values c^{nom}_l, $c^{\text{nom}}_{s\text{L}}$, and $c^{\text{nom}}_{s\text{T}}$ from Table I with $z_{\text{err}} = x_{\text{err}} = 0$, and 2) using the estimated values from Table II for c_l, $c_{s\text{L}}$, $c_{s\text{T}}$, z_{err}, and x_{err}. Specifically, we show L-L, TT-T, and T-T

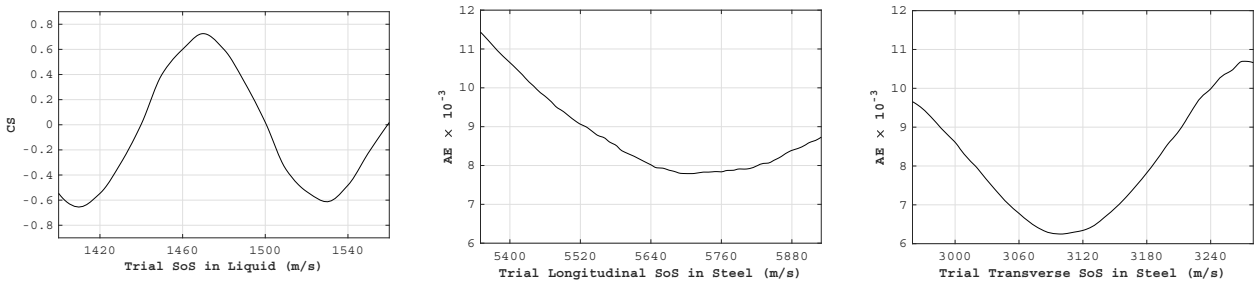


Fig. 4. Cosine similarity and absolute error curves computed during estimation of c_l (left), $c_{s\mathrm{L}}$ (center), and $c_{s\mathrm{T}}$ (right) for sample P_1.

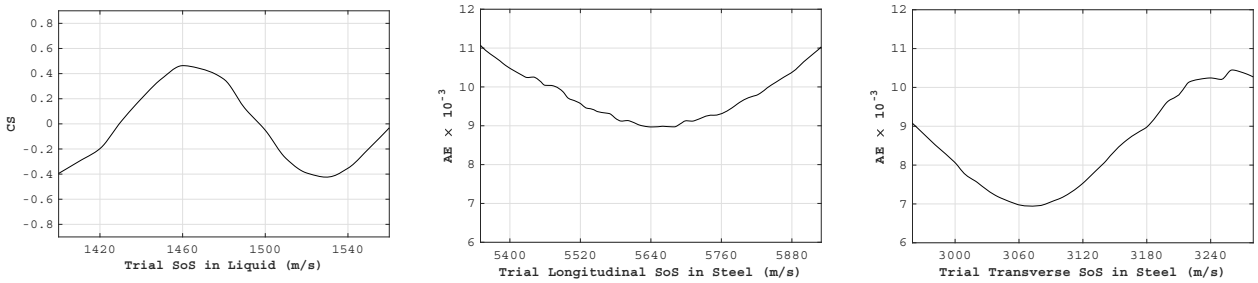


Fig. 5. Cosine similarity and absolute error curves computed during estimation of c_l (left), $c_{s\mathrm{L}}$ (center), and $c_{s\mathrm{T}}$ (right) for sample P_2.

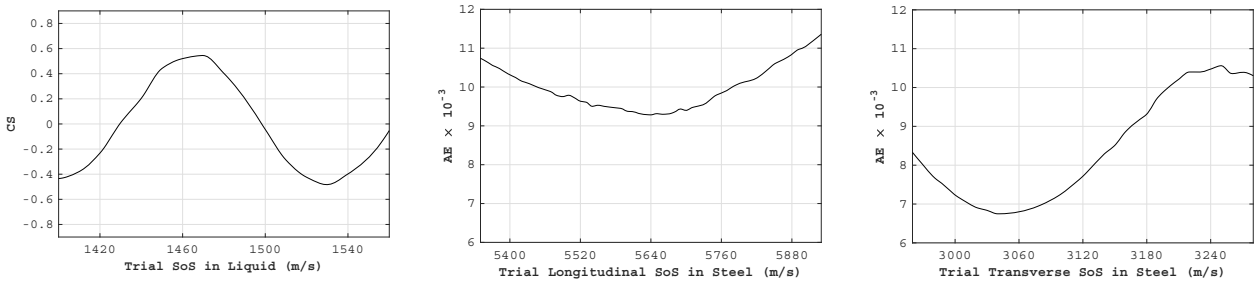


Fig. 6. Cosine similarity and absolute error curves computed during estimation of c_l (left), $c_{s\mathrm{L}}$ (center), and $c_{s\mathrm{T}}$ (right) for sample P_3.

TABLE II
ESTIMATION RESULTS

Pipe Sample	c_l (m/s)	$c_{s\mathrm{T}}$ (m/s)	$c_{s\mathrm{L}}$ (m/s)	z_{err} (mm)	x_{err} (mm)
P_1	1470	3100	5710	2.04	−0.23
P_2	1460	3070	5640	−0.17	−0.77
P_3	1470	3040	5640	0.00	−2.06

mode images of the steel layer, reconstructed from the channel data $\mathcal{A}_0$, $\mathcal{A}_+$, and $\mathcal{A}_-$ for DW emissions at $0°$, $20°$, and $-20°$, respectively.

For sample P_1 in Fig. 7 (cf. Fig. 2, top), the images obtained without SoS and misalignment estimation are obviously worse than those obtained using the speed-of-sound and positioning error values from Table II. In the second scenario, one can see expected visual features of the imaged 5-mm backwall crack, such as its tip in the L-L mode, a vertical line segment in the TT-T mode, and a tilted line segment in the T-T mode.

For sample P_2 in Fig. 8 (cf. Fig. 2, middle), the images in both scenarios are relatively close in terms of their overall visual appearance. In this case, $c_{s\mathrm{L}}$ matches $c_{s\mathrm{L}}^{\mathrm{nom}}$ (see Table I), estimated c_l and $c_{s\mathrm{T}}$ are within 2% of their nominal values, while neither z_{err} nor x_{err} is large enough to significantly impact the image quality.

For sample P_3 in Fig. 9 (cf. Fig. 2, bottom), the displayed L-L mode images without and with SoS and misalignment estimation are clearly different. Given that $c_l \approx 0.99 \cdot c_l^{\mathrm{nom}}$, $c_{s\mathrm{L}} = c_{s\mathrm{L}}^{\mathrm{nom}}$, and $z_{\mathrm{err}} = 0$ (see Table II), this difference is directly attributable to x_{err}. The L-L mode image is certainly better in the second scenario (accounting for x_{err}), as the pipe surface boundaries and two crack tip features are reconstructed more accurately. Meanwhile, the TT-T and T-T mode images in Fig. 9 are somewhat similar across both scenarios. However, if we proceed with the first scenario (assuming $x_{\mathrm{err}} = 0$), the rightmost 4-mm crack would be undersized during subsequent image analysis [10], as its TT-T feature appears substantially smaller than expected.

In closing, we should note that the main disadvantage of our proposed method is the computational cost of reconstructing multiple limited-range trial images $\mathcal{I}_0^*$, $\mathcal{I}_+^*$, and $\mathcal{I}_-^*$ during sound speed and probe misalignment estimation. A cheaper alternative would be to use channel data directly (e.g., see [11], [7]), which is the subject of our future work.

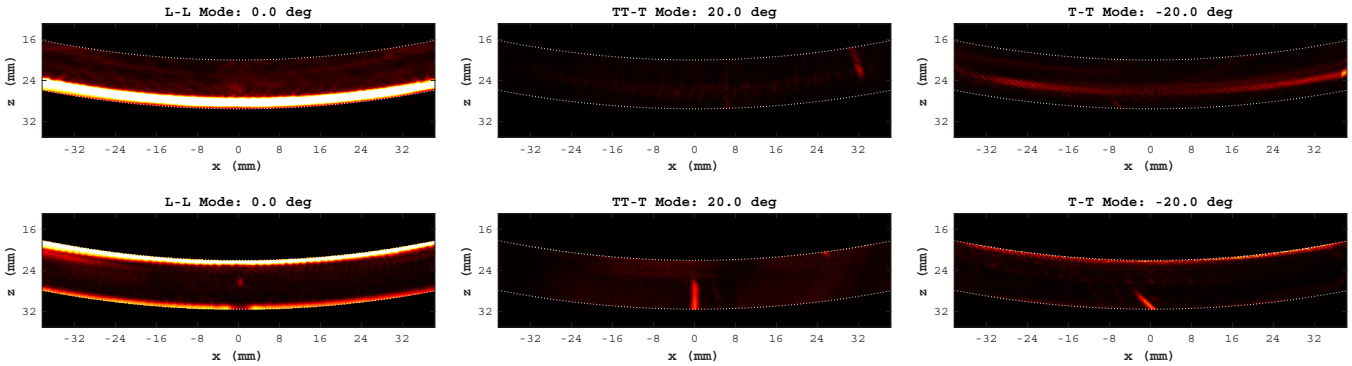


Fig. 7. Examples of reconstructed images without (top) and with (bottom) SoS and misalignment estimation for sample P_1.

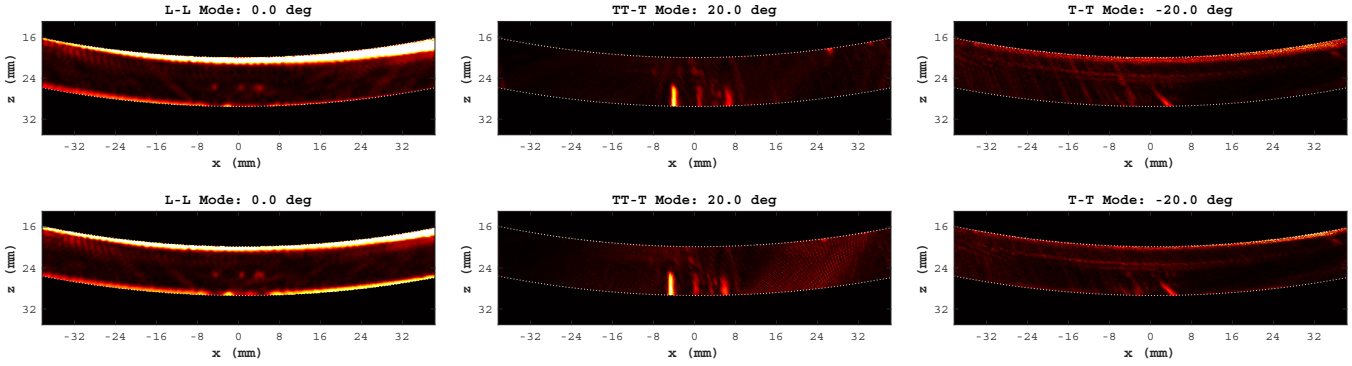


Fig. 8. Examples of reconstructed images without (top) and with (bottom) SoS and misalignment estimation for sample P_2.

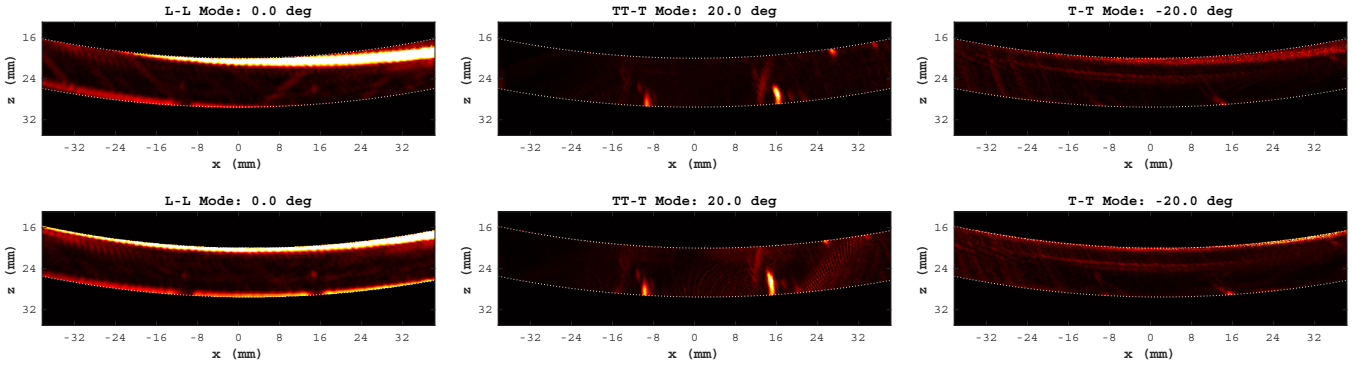


Fig. 9. Examples of reconstructed images without (top) and with (bottom) SoS and misalignment estimation for sample P_3.

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